

# FAIR Data Science

## Part 3

### Context

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In Parts 1 and 2, you defined an individual use case and wrote a first data management plan (DMP). From Part 3 onwards, you work in groups of four. Each group focuses on one use case chosen from the members' Part 1 submissions and develops it into a fully documented, published, and standards-compliant open-science experiment.

This document lists every action you must complete. Each action has a primary owner (A, B, C, or D) who is responsible for the respective section and will be graded for it. During the final session, the instructors will ask each group to demonstrate and explain selected actions. In this session, however, every group member must be prepared to explain any part of the group's work, hence while the work should be distributed across the group members, you must ensure that you also understand the activities performed and results produced by the other group members.

Due to the tight schedule of this exercise, late submissions cannot be accepted. All deadlines are firm. Submissions must be uploaded to TUWEL by the specified deadline. Otherwise, no points will be awarded for the respective part.

Please ensure that all submitted files strictly meet the stated requirements. For example, if a text file requiring a URL to be listed contains more content than the requested URL, no points will be awarded.

As you have learned in previous parts of the course, this exercise involves the use of additional systems in which user accounts may need to be activated. Please take this into account and plan accordingly to ensure that you can complete and submit your work on time.

You are expected to work consistently throughout the entire exercise period. It will not be possible to complete all tasks last minute. We strongly recommend that you avoid working strictly sequentially. For example, do not wait too long to start WP5, as later work packages may depend on earlier preparations.

To successfully pass this exercise, you must achieve at least 40 percent of the available points in each work package.

### Key Platforms

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#### Accounts are required on all platforms before starting

GitHub: <https://github.com>

Zenodo: <https://zenodo.org>

TU Wien Research Data Repository (test instance): <https://test.researchdata.tuwien.ac.at>

DBRepo: <https://www.ifs.tuwien.ac.at/infrastructures/dbrepo/1.13/>

TU Wien DMP Tool: <https://dmptool.tuwien.ac.at>

DaSt-2026-final: <https://test.researchdata.tuwien.ac.at/communities/dast-2026-final> (**not** the one we used in Part 2)

OSTrails shared test repository: <https://gitlab.tuwien.ac.at/crdm/data-stewardship/dast-2026-ue>

RDA maDMP: <https://github.com/RDA-DMP-Common/RDA-DMP-Common-Standard>

OSTrails AP: <https://docs.ostrails.eu/en/latest/commons/dmp/dmp-commons.html>

## Actions

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The list below covers every deliverable. 'Owner' is the student with primary responsibility. Sub-bullets specify mandatory constraints and clarifications.

### WP1 – Preparation

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#### T1.1 Assign Roles [Owner: A, B, C, D]

Form a group of four, label yourselves A–D, and register the group in TUWEL. Grading will be individually based on the individual tasks according to the student(s) primarily responsible.

#### T1.2 Select Use Case [Owner: A, B, C, D]

Agree on which use case from Part 1 the group will use and record this in your TUWEL group page. Identify the use case that seems most interesting to work on, giving preference to use cases that have non-trivial data, with attributes that have semantic meaning (e.g. not just pixel histograms extracted from images; the dataset chosen must contain specific measurements that can be described by some measurement unit) and where you do have a certain competence to understand the features.

#### T1.3 Set-up GitHub [Owner: C]

Create a public GitHub repository for the experiment. Structure it with clear top-level folders (e.g. data/, src/, outputs/, docs/). Apply a consistent, documented file-naming convention described in the README. The naming convention must cover: input datasets, output files (figures, model artefacts), scripts, and configuration files. Document the convention in README.md under a dedicated 'File organisation' section. You will be updating this repository throughout the project, e.g. when reimplementing the experiment or adding more metadata.

#### T1.4 Make a release and update TUWEL [Owner: D]

Make the first release of your project on GitHub, using the release functionality of GitHub. Then upload a TXT file to TUWEL containing the assignment of students to roles and the link to the GitHub release. The file must have 5 rows and be formatted like this:

```
A,family_name,first_name,studentid
B,family_name,first_name,studentid
C,family_name,first_name,studentid
D,family_name,first_name,studentid
LINK to the GitHub RELEASE
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### WP2 – Data Infrastructure

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#### T2.1 DBRepo – schema design [Owner: A]

Design a relational database schema for the experiment's input data in Third Normal Form (3NF). Provide in your repository SQL CREATE statements and an entity–relationship diagram. Create the tables in DBRepo from a Jupyter notebook using the REST API, adding the descriptive metadata for the database and tables to make it citable. Note: make sure the metadata is semantically correct – e.g. be careful not to put TU Wien or yourself as publisher of the data, as you are only re-publishing it, so make sure that along yourself you also point to the original publisher and potential rights owner of the data, ensure proper documentation of license information according to the license information of the original data, etc.

#### T2.2 Semantic mapping [Owner: B]

To address the F and I aspects of FAIR, describe – to the extent possible – all attributes by mapping them to a semantic concept in an ontology: state the ontologies used and justify the choice in 2–3 sentences. Try to select a prominent (i.e. widely used, domain-specific) ontology; refrain from using WikiData unless no other more specific and properly curated ontology can be found. Mappings to top-level or domain-

independent concepts (e.g., schema.org, Dublin Core generic terms) are insufficient unless no domain-specific ontology exists. Add the mapping (via your notebook and the REST API) to the DBRepo metadata.

### T2.3 Mapping units of measurement *[Owner: C]*

For every numeric attribute, map the unit of measurement to an ontological concept. The recommended ontology is the SI Digital Framework:

Documentation:

[https://github.com/TheBIPM/SI\\_Digital\\_Framework/blob/main/SI\\_Reference\\_Point/docs/README.md](https://github.com/TheBIPM/SI_Digital_Framework/blob/main/SI_Reference_Point/docs/README.md)

Specification: <https://si-digital-framework.org/>

SI quantities: <https://si-digital-framework.org/SI>

Serialisation:

[https://github.com/TheBIPM/SI\\_Digital\\_Framework/blob/main/SI\\_Reference\\_Point/TTL/si.ttl](https://github.com/TheBIPM/SI_Digital_Framework/blob/main/SI_Reference_Point/TTL/si.ttl)

In absence of a suitable concept, the fall-back ontology is Quantities and Units

(<https://www.omg.org/spec/Commons>; serialisation:

<https://www.omg.org/spec/Commons/QuantitiesAndUnits.ttl>). If other ontologies are needed, select a suitable one and justify the selection. Preference should again be given to properly curated ontologies. Add the mapping (via your notebook and the REST API) to the DBRepo metadata.

### T2.4 View definitions *[Owner: D]*

De-normalise the data into accessible formats by providing SQL VIEW definitions (descriptively named, not view1/view2) that expose query-ready aggregations useful for the ML pipeline (e.g. feature tables, class-balanced samples). Document each view's purpose in the README.

### T2.5 DBRepo – load *[Owner: C]*

Verify whether the schema defined in T2.1 correctly models the data in 3NF. If so, prepare code to load the input data into DBRepo using the REST API and verify that all views defined in T2.4 return correct results.

### T2.6 DBRepo – API reimplementations *[Owner: D]*

Rewrite the experiment's data-loading code to retrieve data exclusively from the DBRepo REST API instead of local files. This is a mandatory reimplementations – no local CSV/file reads are permitted in the final experiment code. Use the VIEW endpoints wherever applicable. Include robust error handling (connection errors, unexpected response codes). Document the API base URL, all endpoints used, and the authentication method (if any) in the README. Verify that the reimplemented experiment produces results identical to those from the original local-file version and state this explicitly in the final report.

### T2.7 Make a release *[Owner: B]*

Make the second release of your project on GitHub that incorporates all the changes done in WP2. Make sure the release notes cover all the changes from the previous one.

## WP3 – Documentation

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### T3.1 RO-Crate *[Owner: A]*

Create a valid ro-crate-metadata.json file that describes the entire experiment package: input datasets (with source DOIs, if exist), code, trained models (with TUWRD DOI), output data (with TUWRD DOI), authors, licences, and all relationships between entities. Place it in the repository root and reference it in the README. The RO-Crate must pass validation using ro-crate-validator. Include the validation output in the repository under docs/validation/.

### T3.2 CodeMeta *[Owner: B]*

Add a codemeta.json file to the repository root following the CodeMeta 2.0 schema. Must include: name, version, authors (with ORCIDs), license (SPDX identifier), programming language, runtime requirements, all dependencies (with versions), and the GitHub URL as codeRepository. The dependencies list must be

generated programmatically (e.g., from requirements.txt or environment.yml) and must include version pins.

### T3.3 FAIR4ML *[Owner: C]*

Produce FAIR4ML-compliant metadata for each trained ML model. Must include: reference to training dataset (with DOI), algorithm name and version, all hyperparameters, evaluation metrics and their values, intended use, and known limitations. Upload the FAIR4ML metadata file alongside the model in the TUWRD deposit (T3.9) and reference it in the RO-Crate (T3.1).

### T3.4 Croissant *[Owner: D]*

Produce a [Croissant](#) JSON-LD metadata record for each input dataset used in the experiment. The record must describe: fields and their data types, units (referencing QUDT URIs from T2.3), distribution information, and licence. Add it to the repository.

### T3.5 Model Card *[Owner: A]*

Write a Model Card (Markdown file at docs/model-card.md) for each trained model. Required sections: model description, intended use, out-of-scope uses, training data (with DOI), evaluation results (with metrics table), limitations, ethical considerations, and licence. Each required section must be at least 3 sentences. The evaluation results section must include a markdown table with at least precision, recall, and F1 (or equivalent metrics for non-classification tasks). Reference the Model Card from the FAIR4ML metadata (T3.3) and the RO-Crate (T3.1).

### T3.6 Licences *[Owner: B]*

Assign licences explicitly to all three categories of artefact in this experiment – they are not the same and must be considered separately:

**Input data:** identify the licence of the source dataset from Part 1. Verify that your intended use is permitted. State the licence in the README and note any obligations it imposes on derived works (e.g. ShareAlike clauses that affect the output data licence).

**Software / code:** choose an appropriate open-source licence (e.g. MIT, Apache 2.0). Add a LICENSE file to the repository root. Justify the choice in the README, including why it is compatible with the input data licence.

**Produced / output data** (trained models, generated datasets, figures): choose an appropriate open data licence (e.g. CC BY 4.0, CC BY-SA 4.0). State the licence in the README and in every deposit record.

### T3.7 README *[Owner: C]*

Extend README.md to thoroughly cover: project title and abstract, requirements and installation, step-by-step reproduction instructions, description of all inputs and outputs, all three licences (see T3.6), contributors with ORCIDs, and a Zenodo DOI badge (added after T3.8).

### T3.8 Zenodo DOI *[Owner: D]*

Connect the GitHub repository to Zenodo using the GitHub–Zenodo integration. Create a release to mint a DOI. Add the Zenodo DOI badge to the README. Add a CITATION.cff file to the repository root that references the Zenodo DOI. Add the Zenodo DOI as a related identifier in all TU Wien Research Data Repository deposit records.

The Zenodo deposit must be linked to the GitHub repository via the GitHub–Zenodo integration (verifiable by the 'GitHub' source label on the Zenodo record). Manual deposits to Zenodo are not accepted.

### T3.9 Model deposit – TUWRD *[Owner: A]*

Upload the trained ML model artefacts to the TU Wien Research Data Repository test instance. This is a separate deposit from the generated data deposit (T3.10). Provide extensive metadata: title, description, keywords, creators (with ORCIDs), licence (the output-data licence from T3.6), related identifiers (Zenodo DOI, GitHub URL, DBRepo entry URL, generated-data deposit DOI from T3.10). Set the correct resource

type (Model) and add to DaSt-2026-final community. It is enough to request the inclusion in the community, by the time of submission the requests do not have to be approved by the community owners.

### **T3.10 Generated data deposit – TUWRD** [Owner: B]

Upload the generated/output data (predictions, classification results, evaluation figures, confusion matrices, etc.) to the TU Wien Research Data Repository test instance as a separate deposit from the model deposit (T3.9). Provide extensive metadata: title, description, keywords, creators (with ORCIDs), licence, related identifiers (Zenodo DOI, GitHub URL, DBRepo entry URL, model deposit DOI from T3.9). Set the correct resource type (Dataset) and add to DaSt-2026-final community. It is enough to request the inclusion in the community, by the time of submission the requests do not have to be approved by the community owners.

### **T3.11 Standards overlap analysis** [Owner: C]

Analyse the overlap and complementarity between the metadata standards used in this project (RO-Crate, CodeMeta, FAIR4ML, Croissant, Model Card). For each pair of standards, identify: (a) fields that appear in both, (b) fields unique to each, and (c) any conflicts or inconsistencies. Summarise the analysis in a table and discuss the results in the final report. C is responsible for preparing this analysis. All others contribute.

## **WP4 – Final DMP**

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### **T4.1 Write DMP using TU Wien DMP Tool** [Owner: C]

Write the final DMP using TU Wien DMP Tool following the FWF template. All four group members must be listed as contributors in DAMAP. Every field in DAMAP must be completed – no field may be left empty or filled with a placeholder. Note the difference between the final and the initial DMP. Now you describe what you did and can provide detailed information. When listing datasets, group them logically, i.e. do not list every single file.

### **T4.2 Export and finalize DMP** [Owner: D]

Export the DMP as Word document. The exported DMP contains some placeholders and boilerplate text that need to be adapted. Improve the DMP by adding missing information, but do not replace entirely the contents. Do not overpromise or overstate. Save the final DMP as a PDF/A.

### **T4.3 Export and finalize maDMP** [Owner: B]

Export the maDMP from TU Wien DMP Tool as a JSON document. Add missing information to the maDMP based on your group's work (you should use the DMP from T4.2 for reference). Add this information in accordance with the RDA recommendation for maDMPs. You can also use fields from the application profile provided by OSTrails.

### **T4.4 DAMAP – export & publish** [Owner: A]

Submit both the JSON document (T4.3) and the finalized DMP from task 4.2 as a single record in the dedicated DaSt-2026-final community. The deposit must contain two files: the PDF (filename: DMP-FINAL\_<project-title>.pdf) and the maDMP JSON file following the RDA DMP Common Standard schema (filename: maDMP-FINAL\_<project-title>.json). Deposit title: 'DMP-FINAL: <project title>'. Resource type: Data Management Plan. Provide full metadata in the record: all four group members as creators (with ORCIDs), licence (CC BY 4.0), related identifiers pointing to GitHub, TUWRD deposits, and DBRepo entry. Provide also a cross reference in metadata to initial DMP submitted in Part 2 by one of you. Use the "Continues" relation for this. Submit a text file <group\_id>-final-dmp-doi.txt to TUWEL containing the single DOI URL.

### **T4.5 Compare against the initial DMP** [Owner: B]

Compare your final DMP against the initial DMP created by one of you in Part 2. Answer these two questions:

1. What did the initial DMP promise that turned out to be unrealistic, and why? Focus on claims that were never achievable. Do not focus on changes you were asked to make, but things that would not have happened regardless.
2. Looking at the final DMP, what did writing it actually require that the initial one glossed over? What effort, decisions, or knowledge went into the real thing that the initial version skipped?

Include your comparison in the final report. Provide link to both initial and final DMP next to your comparison in the final report.

B is responsible for preparing this discussion. All others contribute.

## WP5 – DMP Evaluation Automation

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### T5.1 Fork and make a pull request *[Owner: A]*

Fork the shared test repository. At the end of Part 5 (all tests contributed), open a pull request from your branch to the upstream repository and name it: “group-<number>”. There needs to be one pull request for the whole group – no individual pull requests to the shared test repository.

### T5.2 Implement tests A *[Owner: A]*

Follow the instructions from the README of the forked repository. Implement tests for the 3 DMP Evaluation Metrics assigned to you – see separate file in TUWEL. Commit them to the forked repository.

### T5.3 Implement tests B *[Owner: B]*

Follow the instructions from the README of the forked repository. Implement tests for the 3 DMP Evaluation Metrics assigned to you – see separate file in TUWEL. Commit them to the forked repository.

### T5.4 Implement tests C *[Owner: C]*

Follow the instructions from the README of the forked repository. Implement tests for the 3 DMP Evaluation Metrics assigned to you – see separate file in TUWEL. Commit them to the forked repository.

### T5.5 Implement tests D *[Owner: D]*

Follow the instructions from the README of the forked repository. Implement tests for the 3 DMP Evaluation Metrics assigned to you – see separate file in TUWEL. Commit them to the forked repository.

## WP6 – Review

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### T6.1 Review – verify artefacts *[Owner: A, B, C, D]*

After the WP4 submission deadline, you will be assigned another group's work to review. Start by verifying whether the actions they described in their DMP were actually implemented: follow every linked persistent identifier and check that the artefacts exist and are as described. Note: you may receive manipulated versions of your colleagues' artefacts, so in your checks do not merely reproduce the content they claim – you need to verify (audit) them.

Check the following:

- GitHub repository (structure, README, licences, file-naming convention, CITATION.cff, Zenodo badge).
- Zenodo deposit (DOI resolves, metadata complete, both licence files present).
- TUWRD deposits for model and generated data (two separate records, metadata, correct resource types, cross-links).
- DBRepo entry (schema, views, semantic annotations visible).
- DMP record (both PDF and maDMP present, maDMP validates against the RDA schema, how they differ).

### T6.2 Review – write report *[Owner: A, B, C, D]*

Write a structured review (PDF, max 3 pages) of the other group's DMP and published artefacts. For each dimension below, provide 2–4 sentences and a rating (1 = poor, 2 = adequate, 3 = good):

- (a) completeness of all FWF DMP fields;
- (b) quality and specificity of data descriptions;
- (c) correctness of metadata standard artefacts (RO-Crate, CodeMeta, FAIR4ML, Croissant, Model Card);
- (d) maDMP technical validity;
- (e) consistency between DMP text and published artefacts;

Submit as <group\_id>-review-of-<other\_group\_id>.pdf to TUWEL.

## WP7 – Reporting

### T7.1 Final report [Owner: A,B,C,D]

Coordinate assembly and submission of the final group report (PDF, submitted to TUWEL as <group\_id>-final-report.pdf).

Required sections:

- (1) title / authors
- (2) abstract (short use case description)
- (3) DBRepo schema, ER diagram, ontology and unit mapping table
- (4) standards application and overlap analysis (T3.11)
- (5) DMP Comparison (T4.5)
- (6) OSTRails test results table (metric ID, category, outcome, message) for your own maDMP.
- (7) individual contribution and sign-off tables (one per student, see template below).

Individual contribution and sign-off tables (one per student). Each student fills in their own rows (columns 1–4); the confirming student fills in columns 5–6. Confirmation chain: A confirmed by B – B confirmed by C – C confirmed by D – D confirmed by A. Copy this template four times, once per student.

Student [A/B/C/D] — [Full Name] — Matriculation No.: \_\_\_\_\_

| Task        | What I did (2–4 sentences)                         | Link / DOI / commit | Done? (✓ / X) | How it was checked (by confirming student)                                           | Confirmed by (A/B/C/D) |
|-------------|----------------------------------------------------|---------------------|---------------|--------------------------------------------------------------------------------------|------------------------|
| <b>Tx.x</b> | Describe what you did and what decisions you made. | https://...         | ✓ / X         | Describe what was verified and how (e.g. followed DOI, re-ran code, checked schema). |                        |
| <b>Tx.x</b> | Describe what you did and what decisions you made. | https://...         | ✓ / X         | Describe what was verified and how (e.g. followed DOI, re-ran code, checked schema). |                        |
| ...         |                                                    |                     |               |                                                                                      |                        |

Add rows as needed. The shaded columns (“How it was checked” and “Confirmed by”) are filled in by the confirming student, not the owner. Repeat this table for each of the four students.

## Deadlines

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30.04 – group registration in TUWEL closes  
07.05 - WP1 submitted  
30.05 – Final DMP submitted  
15.06 – Link to pull request submitted  
22.06 – DMP reviews submitted and final report submitted  
Last week of June – face to face meetings with groups - TBD